

KUNAL MATHUR

Phone: +91-7597209058 | Email: mathurkunal000@gmail.com | LinkedIn: linkedin.com/in/kunal-mathur-612108267
GitHub: github.com/kunal202426 | Portfolio: portfolio-kunal26.vercel.app

SUMMARY

Full Stack SDE at YES Securities (YES Bank), shipping production features on the OMNI trading platform handling 20,000+ daily trades. Strong in React, Node.js, and PostgreSQL at scale, with applied ML using LSTM, LightGBM, and FinBERT ensembles. Overall winner at IIT Madras Global Hyperloop Competition and national semifinalist at Flipkart GRiD 7.0.

EDUCATION

Vellore Institute of Technology
B.Tech, Computer Science & Engineering

2022 – 2026
Vellore, TN

EXPERIENCE

YES Securities (YES Bank)

Jan 2026 – Jun 2026

Full Stack Developer Intern

Bengaluru, India

- Shipped 5+ production features on the OMNI trading application (React, Node.js, RESTful APIs) across following Agile methodologies (sprints, code reviews), supporting a platform that processes 20,000+ daily trades; refactored backend microservices to cut average API latency by ~25%.
- Diagnosed N+1 query bottlenecks and redesigned PostgreSQL schemas, reducing critical query execution time by ~35%; streamlined CI/CD pipelines across 3 backend microservices to accelerate release cadence in a regulated financial environment.

PROJECTS

Distributed Systems Design Simulator

Oct 2025 – Nov 2025

- Built an interactive web platform (React, ReactFlow, Node.js) to design, visualize, and stress-test distributed system architectures in real time; modeled failure modes via M/M/1 queueing theory, chaos engineering, and probabilistic fault injection to surface latency, throughput, and bottleneck metrics across simulated request loads.

CityFlow: Real-Time 3D City Analytics Platform

Feb 2026 – Apr 2026

- Built a real-time interactive 3D city simulation (Three.js, React) modeling urban logistics for vehicles across 5 demand variables (fleet size, surge pricing, weather, demand zones, time-of-day), with a live analytics dashboard tracking 4 KPIs via WebSocket-driven updates for interactive what-if exploration.

Hybrid Stock Price Prediction System

Jan 2026 – Apr 2026

- Engineered an end-to-end ML forecasting ensemble (LSTM + LightGBM + FinBERT sentiment) with ETL pipelines for OHLCV market data and financial news, deployed via FastAPI; reduced MAE by ~18% over single-model baselines. Independent project alongside YES Bank fintech internship.

ACHIEVEMENTS

Winner: Global Hyperloop Competition (Asia's First), IIT Madras

2025

- As AI and Software Lead at Team Levitate (VIT Vellore), built the team website with real-time 3D pod simulations (Three.js, React) and hardware telemetry pipelines achieving sub-50ms data refresh; contributed to 1 overall team trophy and 5 category awards out of 15+ competing international teams.

National Semifinalist: Flipkart GRiD 7.0

2025

- Advanced to national semifinals from thousands of participants in Flipkart's flagship software engineering competition.

CERTIFICATIONS

Data Analytics: *GeeksforGeeks, 2026* · 98% score (874/895) in Python, Pandas, NumPy, SQL, Power BI, Tableau

Blockchain Developer: *IBM Career Education, 2025* · Smart contracts, Hyperledger Fabric, DApp architecture

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frontend & Backend: React.js, Next.js, Three.js, ReactFlow, Node.js, Express.js, FastAPI, RESTful APIs, WebSockets

Databases, DevOps & Tools: PostgreSQL, MongoDB, Redis, Firebase, CI/CD, Figma